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Support bracket, sub-backlight module, display device and splicing display module

Abstract

Provided is a support bracket for supporting a diffuser, including: a support post and a mounting part arranged in a first direction. The mounting part includes: an elastic connection part and a base arranged opposite to each other along the first direction, the elastic connection part is connected to the base, the support post extends in the first direction and an end of the support post close to the mounting part is fixed to the elastic connection part, and the elastic connection part is configured to be elastically deformable in the first direction to change a distance between the end of the support post close to the mounting part and the base in the first direction. Further provided are a sub-backlight module, a display device and a splicing display module.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to the field of display technology, and in particular relates to a support bracket, a sub-backlight module, a display device and a splicing display module.

BACKGROUND

(2) The direct type backlight technology involves placing a backlight module directly below a liquid crystal panel, which can help to obtain more uniform backlight brightness of a liquid crystal television, as well as finer and more vivid images.

(3) In the conventional direct type backlight module, all sides of the diffuser are supported by an intermediate frame and the diffuser is positioned above the back plate. However, the diffuser is relatively thin, leaving a large clearance with the back plate. In order to prevent the diffuser from being recessed in the middle and thus affecting the display image, a support bracket is typically provided in the middle of the diffuser to avoid deformation of the diffuser. However, it is found in practical applications that collision between the diffuser and the support bracket inevitably occurs during installation and use of the diffuser, which may easily cause damages to the diffuser.

SUMMARY

(4) To solve at least one of the technical problems in the existing art, the present disclosure provides a support bracket, a sub-backlight module, a display device and a splicing display module.

(5) In a first aspect, an embodiment of the present disclosure provides a support bracket for supporting a diffuser, including: a support post and a mounting part arranged in a first direction, wherein the mounting part includes: an elastic connection part and a base arranged opposite to each other along the first direction, the elastic connection part is connected to the base, the support post extends in the first direction and an end of the support post close to the mounting part is fixed to the elastic connection part, and the elastic connection part is configured to be elastically deformable in the first direction to change a distance between the end of the support post close to the mounting part and the base in the first direction.

(6) In some embodiments, the elastic connection part includes: two elastic parts and a connection part arranged in a second direction, wherein the connection part is positioned between the two elastic parts, one end of each elastic part is connected to the base, the other end of the elastic part is connected to the connection part, and the end of the support post close to the mounting part is fixed to the connection part; and each elastic part is configured to be elastically deformable in the first direction to change a distance between the connection part and the base in the first direction, wherein the first direction is intersected with the second direction.

(7) In some embodiments, the elastic part is outwardly convex toward a side away from the base, and the connection part is arranged parallel to the base.

(8) In some embodiments, the elastic part includes: a first part, a second part and a third part connected in sequence, wherein the first part is connected to the base, and the third part is connected to the connection part; from the connection between the second part and the first part to the connection between the second part and the third part, a vertical distance between the second part and the base in the first direction gradually increases along the second direction; and from the connection between the third part and the second part to the connection between the third part and the connection part, a vertical distance between the third part and the base in the first direction gradually decreases along the second direction.

(9) In some embodiments, on the support post away from the mounting part in the first direction, the support post has a cross section perpendicular to the first direction with an area that first remains constant and then gradually decreases.
(10) In some embodiments, the support post has the cross section perpendicular to the first direction with a rectangular or trapezoidal shape.
(11) In some embodiments, on the support post away from the mounting part in the first direction, the support post has a cross section perpendicular to the first direction with an area that gradually decreases.
(12) In some embodiments, the support post has the cross section perpendicular to the first direction with a circular shape; the end of the support post close to the mounting part has a cross section perpendicular to the first direction with a radius of 4 mm to 5 mm; an end of the support post away from the mounting part has a cross section perpendicular to the first direction with a radius of 0.5 mm to 1.5 mm; and the support post has a length of 20 mm to 30 mm in the first direction.
(13) In some embodiments, the base is provided with a first screw hole.
(14) In some embodiments, a portion of the base surrounding the first screw hole is outwardly convex toward a side away from the support post.
(15) In some embodiments, the elastic connection part, the base and the support post are integrally formed.
(16) In a second aspect, an embodiment of the present disclosure further provides a sub-backlight module, including: the support bracket as provided in the first aspect.
(17) In some embodiments, the sub-backlight module further includes: a back plate including a bottom plate and a side plate, wherein the side plate is positioned on a first side of the bottom plate and connected to an edge portion of the bottom plate; and
(18) the support bracket is positioned on the first side of bottom plate, with the mounting part of the support bracket fixed to the bottom plate.
(19) In some embodiments, the sub-backlight module further includes: a plurality of light bars arranged in a third direction, wherein each light bar includes a plurality of lamp beads arranged in a fourth direction; and a ratio of the number of lamp beads to the number of support brackets is 6:1 to 10:1.
(20) In some embodiments, for any one of the support brackets, a center of an orthographic projection of the support post on the bottom plate overlaps with a center of a quadrangle enclosed by centers of orthographic projections of four lamp beads closest to the support bracket on the bottom plate.
(21) In some embodiments, a plurality of support brackets arranged in the fourth direction are provided on a central axis of the bottom plate; wherein the central axis of the bottom plate is a virtual line that passes through a center of a surface on a first side of the bottom plate and extends along the fourth direction.
(22) In some embodiments, the surface on the first side of the bottom plate has a first peripheral placement region, a central placement region and a second peripheral placement region sequentially arranged in the third direction; the first peripheral placement region and the second peripheral placement region are each provided with at least two support bracket sets, wherein the at least two support bracket sets include: a first support bracket set, and a second support bracket set on a side of the first support bracket set away from the central placement region, and at least two light bars are provided between the first support bracket set and the second support bracket set; and the first support bracket set and the second support bracket set each include at least two of the support brackets arranged in the fourth direction, and the number of support brackets in the first support bracket set is larger than the number of support brackets in the second support bracket set.
(23) In some embodiments, a reflective sheet is disposed on a side of the light bar facing away from the bottom plate; the reflective sheet is provided with a plurality of first openings in one-to-

one correspondence with the lamp beads, and each lamp bead is exposed out of the corresponding first opening; the reflective sheet is provided with a plurality of second openings in one-to-one correspondence with the support brackets, and the bottom plate is provided with a plurality of second screw holes; and each support bracket is fixed to the bottom plate via a screw penetrating the corresponding second opening, the first screw hole, and the corresponding second screw hole in the bottom plate.

(24) In some embodiments, all the light bars are divided into a plurality of light bar sets, each light bar set includes at least one light bar, each light bar set is configured with a corresponding adapter plate, and each light bar is electrically connected to the corresponding adapter plate through a first flexible flat cable; the adapter plate is fixed on a first side of the bottom plate, and the adapter plate is electrically connected to a power supply control module on a second side of the bottom plate through a second flexible flat cable penetrating the bottom plate; and a receiving slot is formed on the bottom plate in a region corresponding to the adapter plate, and the adapter plate is positioned in the corresponding receiving slot.

(25) In some embodiments, the sub-backlight module further includes: a intermediate frame and a diffuser, wherein the intermediate frame is assembled with and fixed to the back plate; the intermediate frame includes: a first support structure and a second support structure, wherein a side of the first support structure away from the bottom plate is a first support surface, and the second support structure is positioned on a side of the first support surface away from the bottom plate; the diffuser is positioned on the side of the first support surface away from the bottom plate and on an inner side of the second support structure, a clearance space is formed between the diffuser and an inner side wall of the second support structure, and an orthographic projection of the diffuser on the first support surface overlaps the first support surface; a light guide bar is provided between the first support surface and the diffuser; a first side of the light guide bar close to the second support structure is in contact with the inner side wall of the second support structure; and a second side of the light guide bar away from the second support structure exceeds an inner edge of the first support surface.

(26) In some embodiments, a prism film is formed on the inner side wall of the second support structure, and configured to reflect the light in the clearance space emitted to the inner side wall of the second support structure toward the light guide bar.

(27) In some embodiments, the prism film is a Fresnel prism film.

(28) In some embodiments, the intermediate frame includes four support frame bars connected end to end in sequence, each support frame bar includes the first support structure and the second support structure, and a side of the second support structure away from the bottom plate is a second support surface; the four support frame bars include one first support frame bar and three second support frame bars; and a ratio of a width of the second support surface on the first support frame bar to a width of the second support surface on each second support frame bar is 8:1 to 20:1.

(29) In a third aspect, an embodiment of the present disclosure further provides a display device, including: the sub-backlight module as provided in the second aspect.

(30) In some embodiments, the display device further includes: a display panel fixed to the second support surface; b) the display panel includes an active area and a peripheral area surrounding the active area, and the peripheral area includes: one chip-on-film side region and three non-chip-on-film side regions; wherein c) the chip-on-film side region is arranged opposite to the second support surface on the first support frame bar, and the three non-chip-on-film side regions are respectively arranged opposite to the second support surfaces on the three second support frame bars.

(31) In some embodiments, the chip-on-film side region is fixed to the second support surface on the first support frame bar via a double-sided tape; and each non-chip-on-film side region is fixed to the second support surface on the corresponding second support frame bar via ultraviolet curing glue.

(32) In a fourth aspect, an embodiment of the present disclosure further provides a splicing display module, including: 2N display devices, the 2N display devices are arranged in an array of two rows and N columns, where N is a positive integer; chip-on-film side regions of N display devices in a first row are all positioned on a side away from a second row of display devices; and chip-on-film side regions of N display devices in the second row are all positioned on a side away from the first row of display devices.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1**A is a schematic structural view of a support bracket according to an embodiment of the present disclosure;
(2) FIG. **1**B is a schematic sectional view of the support bracket shown in FIG. **1**A;
(3) FIG. **2**A is another schematic structural view of a support bracket according to an embodiment of the present disclosure;
(4) FIG. **2**B is another schematic sectional view of the support bracket shown in FIG. **2**A;
(5) FIG. **3** is a schematic structural view of a sub-backlight module according to an embodiment of the present disclosure;
(6) FIG. **4** is a schematic sectional view taken along line A-A′ of FIG. **3**;
(7) FIG. **5** is a schematic top view of a back plate according to an embodiment of the present disclosure;
(8) FIG. **6** is a schematic structural view of lamp beads in a partial region of a back plate and a support bracket;
(9) FIG. **7** is a schematic structural view of an adapter plate disposed on a back plate according to an embodiment of the present disclosure;
(10) FIG. **8** is a schematic sectional view of a diffuser according to the existing art when an edge portion of the diffuser is placed on a first support surface;
(11) FIG. **9** is a schematic sectional view of a diffuser according to an embodiment of the present disclosure when an edge portion of the diffuser is placed on a first support surface;
(12) FIG. **10** is another schematic sectional view of a diffuser according to an embodiment of the present disclosure when an edge portion of the diffuser is placed on a first support surface;
(13) FIG. **11** is a schematic structural view showing assembly of the intermediate frame and the back plate according to an embodiment of the present disclosure;
(14) FIG. **12**A is a schematic top view of a second support surface of four support frame bars in an intermediate frame according to an embodiment of the present disclosure;
(15) FIG. **12**B is a schematic structural view showing connection between two adjacent support frame bars according to an embodiment of the present disclosure;
(16) FIG. **13** is an exploded structural view of a display device according to an embodiment of the present disclosure;
(17) FIG. **14** is a schematic sectional view of a display device according to an embodiment of the present disclosure;
(18) FIG. **15**A is a schematic structural view of a second support frame bar with a first protective film adhered to an outer side wall of the second support frame bar;
(19) FIG. **15**B is a schematic sectional view of a second support frame bar with a first protective film adhered to an outer side wall of the second support frame bar;
(20) FIG. **16**A is a schematic structural view of a second support frame bar with a masking tape adhered to a second support surface of the second support frame bar;
(21) FIG. **16**B is a schematic sectional view of a second support frame bar with a masking tape adhered to a second support surface of the second support frame bar;

(22) FIG. **17**A is a schematic structural view of a second support frame bar with a first protective film adhered to an inner edge of a second support surface of the second support frame bar;
(23) FIG. **17**B is a schematic sectional view of a second support frame bar with a second protective film adhered to an inner edge of a second support surface of the second support frame bar;
(24) FIG. **18**A is a schematic structural view of a second support frame bar with ultraviolet curing glue coated on a second support surface of the second support frame bar;
(25) FIG. **18**B is a schematic sectional view of a second support frame bar with ultraviolet curing glue coated on a second support surface of the second support frame bar;
(26) FIG. **19** is a schematic structural view when the masking tape is removed;
(27) FIG. **20** is a schematic structural view of placing a support block in a region where the masking tape is removed; and
(28) FIG. **21** is a schematic structural view of a splicing display module according to an embodiment of the present disclosure.

DETAIL DESCRIPTION OF EMBODIMENTS

(29) To better understand the technical solution of the present disclosure for those skilled in the art, the support bracket, the sub-backlight module, the display device and the splicing display module of the present disclosure will be described in detail below in conjunction with the accompanying drawings.
(30) FIG. **1**A is a schematic structural view of a support bracket according to an embodiment of the present disclosure, and FIG. **1**B is a schematic sectional view of the support bracket shown in FIG. **1**. As shown in FIGS. **1**A and **1**, the support bracket is used for supporting a diffuser and includes: a support post **1** and a mounting part **2** arranged in a first direction Y. The mounting part **2** includes: an elastic connection part **3** and a base **4** arranged opposite to each other along the first direction Y. The elastic connection part **3** is connected to the base **4**, and the support post **1** extends in the first direction Y, and an end of the support post **1** close to the mounting part **2** is fixed to the elastic connection part **3**, and the elastic connection part **3** is configured to be elastically deformable in the first direction Y, so as to change a distance between the end of the support post **1** close to the mounting part **2** and the base **4** in the first direction Y.
(31) In an embodiment of the present disclosure, by providing the elastic connection part **3** between the base **4** and the support post **1**, when the diffuser collides with an end of the support post **1** away from the mounting part **2**, the elastic connection part **3** may be elastically deformed under an applied force, thereby having a cushioning effect, and thus effectively reducing the cracking risk of the diffuser.
(32) In some embodiments, the elastic connection part **3** includes: two elastic parts **3***a* and a connection part **3***b* arranged in a second direction X. The connection part is positioned between the two elastic parts **3***a*. One end of each elastic part **3***a* is connected to the base **4**, and the other end of the elastic part **3***a* is connected to the connection part **3***b*. The end of the support post **1** close to the mounting part **2** is fixed to the connection part **3***b*. Each elastic part **3***a* is configured to be elastically deformable in the first direction Y, so as to change a distance between the connection part **3***b* and the base **4** in the first direction Y, where the first direction Y is intersected with the second direction X.
(33) By providing the above two elastic parts **3***a* and the connection part **3***b*, on one hand, stable support of the support post **1** can be realized, and on the other hand, corresponding elastic deformation can be rapidly generated when the support post **1** is subjected to an external force.
(34) In some embodiments, the elastic part **3***a* is outwardly convex toward a side away from the back plate, and the connection part is arranged parallel to the base **4**.
(35) In some embodiments, the elastic part **3***a* has a bar shape and includes: a first part **301**, a second part **302** and a third part **303** connected in sequence. The first part **301** is connected to the base **4**, and the third part **303** is connected to the connection part **3***b*. From the connection between the second part **302** and the first part **301** to the connection between the second part **302** and the

third part **303**, a vertical distance between the second part **302** and the base in the first direction Y gradually increases along the second direction X. From the connection between the third part **303** and the second part **302** to the connection between the third part **303** and the connection part, a vertical distance between the third part **303** and the base in the first direction Y gradually decreases along the second direction X.

(36) It should be noted that the case where the elastic connection part **3** includes two elastic parts **3***a* and one connection part **3***b* is merely an optional implementation of the embodiment of the present disclosure, and does not form any limitation to the technical solution of the present disclosure. In the present disclosure, the elastic connection part **3** may also adopt other structures capable of generating elastic deformation under the action of an external force, which are not listed one by one here.

(37) In some embodiments, when the diffuser is not supported on the support bracket, the connection part **3***b* and the base **4** has a distance h in the first direction Y, and 0.45 mm≤h≤0.55 mm.

(38) Referring to FIGS. **1**A and **1**A, in some embodiments, on the support post **1** away from the mounting part **2** in the first direction Y, the support post **1** has a cross section perpendicular to the first direction Y with an area that first remains constant and then gradually decreases.

(39) In some embodiments, the support post has the cross section perpendicular to the first direction with a rectangular or trapezoidal shape.

(40) Referring to FIGS. **2**A and **2**B, in some embodiments, on the support post **1** away from the mounting part **2** in the first direction Y, the support post **1** has a cross section perpendicular to the first direction Y with an area that gradually decreases. With such a design, the support post **1** presents an overall shape narrow at the top and wide at the bottom, which on one hand reduces an overall weight of the support post **1**, and on the other hand enables a larger connection area between the bottom end (the end close to the mounting part **2**) of the support post **1** and the mounting part **2**, so as to ensure firm connection between the support post **1** and the mounting part **2**. Meanwhile, the top end (the end away from the mounting part **2**) of the support post **1** will not be too large, so that a contact area between the support post **1** and the diffuser is relatively small, which is beneficial to increasing an amount of light incident on the diffuser.

(41) In addition, considering that the smaller a top area of the support post **1** is, the greater the pressure intensity on the diffuser is when the support post **1** collides with the diffuser, and the higher the risk of the diffuser being broken is, the top area of the support post **1** cannot be designed to be too small.

(42) In some embodiments, the support post **1** has the cross section perpendicular to the first direction Y with a circular shape. The end of the support post **1** close to the mounting part **2** has a cross section perpendicular to the first direction Y with a radius of 4 mm to 5 mm. The end of the support post **1** away from the mounting part **2** has a cross section perpendicular to the first direction Y with a radius of 0.5 mm to 1.5 mm. The support post **1** has a length of 20 mm to 30 mm in the first direction Y.

(43) With continued reference to FIGS. **1** and **2**, in some embodiments, the base **4** is provided with a first screw hole **5**. A second screw hole is formed on the back plate at a position corresponding to a region where the support bracket can be placed. When the support bracket is assembled with the back plate, the support bracket can be fixed at a corresponding position on the back plate simply by passing a screw through the first screw hole **5** in the support bracket and the second screw hole in the back plate at the corresponding position.

(44) In some embodiments, a portion **6** of the base **4** surrounding the first screw hole **5** is outwardly convex toward a side away from the support post **1**. A portion of the bottom plate surrounding the second screw hole may be outwardly convex toward a side facing away from the support surface to form a positioning slot. When the support bracket is assembled with the back plate, the outwardly convex portion **6** surrounding the first screw hole **5** on the support bracket may be placed into the

positioning slot in the bottom plate, which can, on one hand, implement alignment of the first screw hole **5** and the second screw hole, and on the other hand, have a function of limiting the support bracket to prevent mismatched movement between the support bracket and the back plate.
(45) In some embodiments, the elastic connection part **3**, the base **4** and the support post **1** are each made of a transparent resin material, which can effective avoid shielding of the light emitted from the backlight source after the support bracket is assembled with the back plate.
(46) In some embodiments, the elastic connection part **3**, the base **4** and the support post **1** are integrally formed. Optionally, the integrally formed elastic connection part **3**, base **4** and support post **1** may be formed by injection molding.
(47) FIG. **3** is a schematic structural view of a sub-backlight module according to an embodiment of the present disclosure, FIG. **4** is a schematic sectional view taken along line A-A′ of FIG. **3**, FIG. **5** is a schematic top view of a back plate according to an embodiment of the present disclosure, and FIG. **6** is a schematic structural view of lamp beads in a partial region of a back plate and a support bracket. As shown in FIGS. **3** to **6**, the sub-backlight module includes a support bracket **19**, which adopts the support bracket **19** provided in the above embodiment. For the description of the support bracket **19**, reference may be made to the contents in the foregoing embodiment, which is not repeated here.
(48) In some embodiments, the sub-backlight module further includes a back plate **12** including a bottom plate **12***a* and a side plate **12***b*. The side plate **12***b* is positioned on a first side of the bottom plate **12***a* and connected to an edge portion of the bottom plate **12***a*. The support bracket **19** is positioned on the first side of bottom plate **12***a*, with the mounting part of the support bracket **19** fixed to the bottom plate **12***a*.
(49) The sub-backlight module further includes a plurality of light bars **15** arranged in a third direction W. Each light bar **15** includes a plurality of lamp beads **16** arranged in a fourth direction Z. The light bars **15** may be fixed on the back plate **12** by a double-sided tape.
(50) In some embodiments, a ratio of the number of lamp beads **16** to the number of support brackets **19** is 6:1 to 10:1. In practical applications, the number and arrangement of the lamp beads **16** may be set according to the brightness requirement of the display device, while the number and arrangement of the support brackets **19** may be set according to the supporting requirement of the diffuser **13**. In addition, with a fixed number of lamp beads **16**, more support brackets **19** can provide a better supporting effect for the diffuser **13**, but with a greater influence on light emission of the light bars **15**. In an embodiment of the present disclosure, comprehensively considering the light emitting effect of the light bars **15** and the support reliability of the diffuser **13**, the ratio of the number of lamp beads **16** to the number of support brackets **19** may be set to the range of 6:1 to 10:1, such as 8:1.
(51) In some embodiments, for any one of the support brackets **19**, a center of an orthographic projection of the support post **1** of the support bracket **19** on the bottom plate **12***a* overlaps with a center of a quadrangle enclosed by centers of orthographic projections of four lamp beads **16** closest to the support bracket **19** on the bottom plate **12***a*.
(52) Referring to FIG. **5**, in some embodiments, a plurality of support brackets **19** arranged in the fourth direction Z are provided on a central axis L of the bottom plate **12***a*. The central axis L of the bottom plate **12***a* is a virtual line that passes through a center of a surface on a first side of the bottom plate **12***a* and extends along the fourth direction Z. In an embodiment of the present disclosure, considering that a middle portion of the diffuser tends to be deformed most due to gravity, a column of support brackets **19** are disposed on the central axis L of the bottom plate **12***a* in the embodiment of the present disclosure.
(53) In some embodiments, the surface on the first side of the bottom plate **12***a* has a first peripheral placement region **12***al*, a central placement region **12***a***2** and a second peripheral placement region **12***a***3** sequentially arranged in the third direction W. The first peripheral placement region **12***al* and the second peripheral placement region **12***a***3** are each provided with at

least two support bracket sets. The at least two support bracket sets include: a first support bracket set **19***a*, and a second support bracket set **19***b* on a side of the first support bracket set **19***a* away from the central placement region **12***a***2**. At least two light bars **15** are provided between the first support bracket set **19***a* and the second support bracket set **19***b*. The first support bracket set **19***a* and the second support bracket set **19***b* each include at least two of the support brackets **19** arranged in the fourth direction Z, and the number of support brackets **19** in the first support bracket set **19***a* is larger than the number of support brackets **19** in the second support bracket set **19***b*.
(54) The drawings merely exemplarily show a case where two light bars **15** are provided between the first support bracket set **19***a* and the second support bracket set **19***b*, the first support bracket set **19***a* includes four support brackets **19**, and the second support bracket set **19***b* includes two support brackets **19**, which is merely exemplary, and does not configure any limitation to the technical solution of the present disclosure.
(55) In some embodiments, the support brackets **19** in the central placement region **12***a***2** are arranged in an array along the third and fourth directions.
(56) Referring to FIG. **5**, the number of the light bars **15** is twenty, and each light bar **15** includes ten lamp beads **16**. Taking the third direction W as a row direction and the fourth direction Z as a column direction, all the lamp beads **16** are arranged in an array of ten rows and twenty columns.
(57) Meanwhile, twenty-four support brackets **19** are provided and specifically distributed as follows: five support brackets **19** are respectively disposed between the second row of lamp beads **16** and the third row of lamp beads **16**, between the fourth row of lamp beads **16** and the fifth row of lamp beads **16**, between the sixth row of lamp beads **16** and the seventh row of lamp beads **16**, and between the eighth row of lamp beads **16** and the ninth row of lamp beads **16**, and the five support brackets **19** are respectively positioned between the fourth column of lamp beads **16** and the fifth column of lamp beads **16**, between the seventh column of lamp beads **16** and the eighth column of lamp beads **16**, between the tenth column of lamp beads **16** and the eleventh column of lamp beads **16**, between the thirteenth column of lamp beads **16** and the fourteenth column of lamp beads **16**, and between the sixteenth column of lamp beads **16** and the seventeenth column of lamp beads **16**; and two support brackets **19** are respectively disposed between the third row of lamp beads **16** and the fourth row of lamp beads **16**, and between the seventh row of lamp beads **16** and the eighth row of lamp beads **16**, and the two support brackets **19** are respectively positioned between the second column of lamp beads **16** and the third column of lamp beads **16**, and between the eighteenth column of lamp beads **16** and the nineteenth column of lamp beads **16**.
(58) Apparently, the number and distribution of the lamp beads **16** and the number and distributed of the support brackets **19** shown in FIG. **5** merely form an optional implementation for the embodiment of the present disclosure, and do not form any limitation to the technical solution of the present disclosure.
(59) In some implementations, a reflective sheet **30** is disposed on a side of the light bar **15** facing away from the bottom plate **12***a*. The reflective sheet **30** is provided with a plurality of first openings in one-to-one correspondence with the lamp beads **16**, and each lamp bead **16** is exposed out of the corresponding first opening. By providing the reflective sheet **30** on the back plate **12**, the light emitted to the bottom plate **12***a* can be reflected to the diffuser **13**, which is beneficial to improving the utilization rate of light.
(60) Further optionally, when the base of the support bracket **19** is provided with a first screw hole, a plurality of second openings in one-to-one correspondence with the support brackets **19** may be provided in the reflective sheet **30**, and a plurality of second screw holes may be provided in the bottom plate **12***a*. Via a screw penetrating the corresponding second opening, the first screw hole in the support bracket **19**, and the corresponding second screw hole in the bottom plate **12***a*, the support bracket **19** is fixed to the bottom plate **12***a*.
(61) FIG. **7** is a schematic structural view of an adapter plate disposed on a back plate according to

an embodiment of the present disclosure. As shown in FIGS. **6** and **7**, in some embodiments, all the light bars **15** are divided into a plurality of light bar **15** sets each including at least one light bar **15**. Each light bar **15** set is configured with a corresponding adapter plate **20**, and each light bar **15** is electrically connected to the corresponding adapter plate **20** through a first flexible flat cable **21**. The adapter plate **20** is fixed on a first side of the bottom plate **12***a*. The adapter plate **20** is electrically connected to a power supply control module **80** (shown in FIG. **13** later) on a second side of the bottom plate **12***a* through a second flexible flat cable **22** penetrating the bottom plate **12***a*.

(62) The power supply control module may provide an electrical signal to the light bar **15** through the second flexible flat cable **22**, the adapter plate **20**, and the first flexible flat cable **21**. In some embodiments, the power supply control module has a local dimming function, which is beneficial to improving the contrast of the display screen.

(63) In some embodiments, a receiving slot **31** is formed on the bottom plate **12***a* in a region corresponding to the adapter plate **20**, and the adapter plate **20** is positioned in the corresponding receiving slot **31**. Specifically, the portion for placing the adapter plate **20** on the bottom plate **12***a* is outwardly convex toward a side facing away from the support surface to form the receiving slot **31**, and the adapter plate **20** is placed into the receiving slot **31** so that the adapter plate **20** sinks, thereby preventing the adapter plate **20** from locally jacking up the light bar **15**.

(64) As a specific example, the twenty light bars **15** shown in FIG. **5** may be configured with four adapter plates **20**, and each adapter plate **20** may be connected to five light bars **15**.

(65) Referring to FIG. **4**, in some embodiments, the sub-backlight module further includes a intermediate frame **11** and a diffuser **13**. The intermediate frame **11** is assembled with and fixed to the back plate **12**. The intermediate frame **11** includes: a first support structure **11***a* and a second support structure **11***b*. A side of the first support structure **11***a* away from the bottom plate **12***a* is a first support surface, and the second support structure **11***b* is positioned on a side of the first support surface away from the bottom plate **12***a*. The diffuser **13** is positioned on the side of the first support surface away from the bottom plate **12***a* and on an inner side of the second support structure **11***b*. A clearance space (an expansion space reserved for installation of the diffuser **13**) is formed between the diffuser **13** and an inner side wall of the second support structure **11***b*. An orthographic projection of the diffuser **13** on the first support surface overlaps the first support surface. In an embodiment of the present disclosure, the first support surface may be used to support an edge portion of the diffuser **13**, while the support bracket **19** may be used to support a middle portion of the diffuser **13**. In some embodiments, a light guide bar **14** is provided between the first support surface and the diffuser **13**.

(66) FIG. **8** is a schematic sectional view of a diffuser according to the existing art when an edge portion of the diffuser is placed on a first support surface. As shown in FIG. **8**, in the existing art, an edge portion of the diffuser **13** is directly lapped on the first support surface. Due to the narrow bezel design of the display product, after the display panel and the sub-backlight module are assembled, the portion of the diffuser **13** lapped on the first support surface has an orthogonal projection on the display panel that will cover an active area (AA) of the display panel. Since the portion of the diffuser **13** lapped on the first support surface has a low light-emitting brightness (referred to as a “low-brightness light-emitting portion”), light received by a region (i.e., an edge portion of the AA) on the display panel opposite to the low-brightness light-emitting portion is reduced, so that the edge portion of the display screen has a low display brightness, causing the so-called “dark edge” problem.

(67) FIG. **9** is a schematic sectional view of a diffuser according to an embodiment of the present disclosure when an edge portion of the diffuser is placed on a first support surface. As shown in FIG. **9**, in order to effectively solve the “dark edge” problem, a light guide bar **14** is provided on the first support surface, and the diffuser **13** is lapped on the light guide bar **14**. When the light emitted from the lamp beads **16** illuminates the light guide bar **14**, the light guide bar **14** may

transmit the light to the edge portion of the diffuser **13**, so that the edge portion of the diffuser **13** has an increased light-emitting brightness, the light received by the edge portion of the AA of the display panel is increased, and the display brightness at the edge portion of the AA of the display panel is increased, thereby effectively improving and even eliminating the “dark edge” problem.

(68) In some embodiments, a first side of the light guide bar **14** close to the second support structure **11***b* is in contact with the inner side wall of the second support structure **11***b*, and a second side of the light guide bar **14** away from the second support structure **11***b* exceeds an inner edge of the first support surface. The farther the second side of the light guide bar **14** away from the second support structure **11***b* exceeds the inner edge of the first support surface, the thicker the light guide bar **14** is, the more light is incident on the light guide bar **14**, and the more beneficial is to increasing the light-emitting brightness at the edge portion of the diffuser **13**. However, as the second side of the light guide bar **14** away from the second support structure **11***b* exceeds the inner edge of the first support surface farther, the center of gravity of the light guide plate may shift toward a direction away from the inner side wall of the second support structure **11***b*, When the center of gravity of the light guide plate shifts to exceed the inner edge of the first support surface, the light guide plate is likely to be tilted, which is unfavorable for assembly of the light guide plate. In addition, as the light guide bar **14** becomes thicker, the assembly space for the diffuser **13** and an optical film **18** is reduced, which is unfavorable for assembly of the diffuser **13** and the optical film **18**.

(69) Considering the light-emitting brightness at the edge portion of the diffuser **13** and the difficulty of assembling the light guide plate/the diffuser **13**/the optical film **18**, a thickness of the light guide bar **14** in the first direction in the embodiment of the present disclosure is set to 1.2 mm to 1.6 mm, for example, 1.4 mm; and the second side of the light guide bar **14** away from the second support structure **11***b* exceeds the inner edge of the first support surface by 1 mm to 2 mm, for example, 1.5 mm.

(70) Since a clearance space **35** is formed between the diffuser **13** and the inner side wall of the second support structure **11***b*, a part of light on the light guide bar **14** will enter the clearance space **35**, where a small part of the light passes through the clearance space **35** at a small angle (an included angle formed by the light and a normal on a surface of the light guide plate), while most light is incident on the inner side wall of the second support structure **11***b* in a direction at a large angle. Since the inner side wall is generally made of a metal material, the surface of the inner side wall has good specular reflectivity, so that the light incident on the surface of the inner side wall is specularly reflected, and then emitted toward the edge portion of the AA of the display panel at a large angle. In this case, the edge portion of the AA of the display panel appears dark in front view and bright in side view.

(71) FIG. **10** is another schematic sectional view of a diffuser according to an embodiment of the present disclosure when an edge portion of the diffuser is placed on a first support surface. As shown in FIG. **10**, in order to relieve the problem that the edge portion of the AA of the display panel appears dark in front view and bright in side view, according to the technical solution of the present disclosure, a prism film **32** is formed on the inner side wall of the second support structure **11***b*. The prism film **32** is configured to reflect the light in the clearance space **35** emitted to the inner side wall of the second support structure **11***b* toward the light guide bar **14**. In other words, the light incident on the inner side wall of the second support structure **11***b* at a large angle are reflected to the light guide bar **14** by the prism film **32**, where in a portion of the light reaching the light guide bar **14** are refracted and then transmitted again in the light guide bar **14**, and finally pass through the clearance space **35** in a direction at a small angle. Through this technical solution, the light exiting the clearance space **35** in a direction at a large angle is reduced, while the light exiting in a direction at a small angle is increased, so as to effectively relieve, and even eliminate, the problem that the edge portion of the AA of the display panel appears dark in front view and bright in side view.

(72) In practical applications, to reserve an assembly error space, the prism film **32** is designed to have an area slightly smaller than an area of a portion on the inner side wall of the second support structure **11***b* that is not in contact with the light guide bar. Apparently, without considering the assembly error space, the prism film **32** may just cover the portion on the inner side wall of the second support structure **11***b* that is not in contact with the light guide bar (the prism film **32** is designed to have an area equal to the area of the portion on the inner side wall of the second support structure **11***b* that is not in contact with the light guide bar).
(73) It should be noted that the size and reflecting angle of each prism on the prism film may be simulated and designed in advance according to the actual dimming requirement, so as to meet the requirement of actual use. In some embodiments, the prism film is a Fresnel prism film.
(74) FIG. **11** is a schematic structural view showing assembly of the intermediate frame and the back plate according to an embodiment of the present disclosure. As shown in FIG. **11**, in some embodiments, the intermediate frame **11** further includes an assembling part **11***c* fixed to the back plate **12**. The assembling part **11***c* is positioned on a side of the second support structure **11***b* away from the second support structure **11***b*. A sink **11***d* is formed on the assembling part **11***c*, and an opening of the sink **11***d* faces away from the first support structure **11***a*. A side plate **12***b* on the back plate **12** includes a first support plate **1201** connected to the bottom plate **12***a*, a second support plate **1202** connected to the second support plate **1201**, and a third support plate **1203** connected to the second support plate **1202**. The first support plate **1201**, the second support plate **1202** and the third support plate **1203** are formed by a same panel through secondary bending. An outer side wall of the sink **11***d* and the third support plate **1203** are each provided with a screw hole, respectively. The back plate **12** is fixed to the intermediate frame **11** by placing the third support plate **1203** into the sink **11***d* and inserting a screw **17** through the screw holes in the outer side wall and the third support plate **1203**. In some embodiments, an end of the third support plate **1203** contacts the bottom plate **12***a* of the sink **11***d*, and an side wall of the assembling part **11***c* enclosing the sink **11***d* and having a screw hole contacts the second support plate **1202**, so as to implement stable assembly of the plate and the intermediate frame **11**.
(75) FIG. **12**A is a schematic top view of a second support surface of four support frame bars in a intermediate frame according to an embodiment of the present disclosure, and FIG. **12**B is a schematic structural view showing connection between two adjacent support frame bars according to an embodiment of the present disclosure. As shown in FIGS. **12**A and **12**B, in some embodiments, the intermediate frame **11** includes four support frame bars **40** connected end to end in sequence. Each support frame bar **40** includes a first support structure **11***a* and a second support structure **11***b*. Two ends of two adjacent support frame bars **40** may be fixed with a corner block.
(76) In some embodiments, the corner block has a first connection bar and a second connection bar perpendicular to each other. The first connection bar and the second connection bar are each provided with a screw hole. The side wall of the assembling part **11***c* enclosing the sink at an end position of the support frame bar is formed with a screw hole. The first connection bar and the second connection bar of the corner block **43** may be placed into the sink **11***d* of two support frame bars to be connected, and a screw **44** is used to penetrate the screw hole at the end position of the support frame bar and the screw hole in the first connection bar/the second connection bar, so that the two support frame bars **40** to be connected are connected with a same corner block.
(77) In some embodiments, the first connection bar and the second connection bar are each provided with a positioning protrusion **45**, and a positioning opening is formed at the end position of the support frame bar **40**. By placing the positioning protrusion **45** on the first connection bar/the second connection bar in the positioning opening, the support frame bar **40** is positioned relative to the first connection bar or the second connection bar, so as to facilitate the subsequent fixing of the support frame bar and the first connection bar or the second connection bar by using the screw **44**.
(78) In some embodiments, a side of the second support structure **11***b* away from the bottom plate **12***a* is a second support surface. The four support frame bars include one first support frame bar **41**

and three second support frame bars **42**. A ratio of a width of the second support surface on the first support frame bar **41** to a width of the second support surface on each second support frame bar **42** is 8:1 to 20:1, for example, 40:3.
(79) In some embodiments, the width of the second support surface on the first support frame bar **41** is 10 mm to 14 mm, for example, 12 mm. The width of the second support surface on the second support frame bar **42** is 0.8 mm to 1.2 mm, for example, 0.9 mm.
(80) In practical applications, the width of the second support surface on each support frame bar may be designed and adjusted according to actual needs.
(81) Based on a same inventive concept, an embodiment of the present disclosure further provides a display device. FIG. **13** is an exploded structural view of a display device according to an embodiment of the present disclosure, and FIG. **14** is a schematic sectional view of a display device according to an embodiment of the present disclosure. As shown in FIGS. **13** and **14**, the display device includes a sub-backlight module, which may adopt the sub-backlight module provided in any of the above embodiments. For specific description of the sub-backlight module, reference may be made to the contents in the foregoing embodiments, and details are not repeated here.
(82) In some embodiments, when the intermediate frame **11** includes four support frame bars, including one first support frame bar **41** and three second support frame bars **42**, the display device further includes: a display panel **51** fixed to the second support surface of each support frame bar **40**. The display panel includes an active area and a peripheral area surrounding the active area. The peripheral area includes: one chip-on-film side region and three non-chip-on-film side regions. The chip-on-film side region is arranged opposite to the second support surface on the first support frame bar **41**, and the three non-chip-on-film side regions are respectively arranged opposite to the second support surfaces on the three second support frame bars **42**.
(83) The chip-on-film side region is used for realizing a chip on film (COF) process. The COF process includes: placing a display drive IC chip into an FPC flat cable, and then folding to a lower side of a screen by means of the properties of the FPC itself. Specifically, a gold bump on the IC chip is bonded with an inner lead of a flexible substrate circuit by thermocompression bonding. Since the space occupied by the IC chip is released, a width of the bezel can be effectively reduced.
(84) In some embodiments, the chip-on-film side region is fixed to the second support surface on the first support frame bar **41** via a double-sided tape **50***a*. Each non-chip-on-film side region is fixed to the second support surface on the corresponding second support frame bar **42** via ultraviolet curing glue **50***b*.
(85) FIG. **15**A is a schematic structural view of a second support frame bar with a first protective film adhered to an outer side wall of the second support frame bar, FIG. **15**B is a schematic sectional view of a second support frame bar with a first protective film adhered to an outer side wall of the second support frame bar, FIG. **16**A is a schematic structural view of a second support frame bar with a masking tape adhered to a second support surface of the second support frame bar, FIG. **16** is a schematic sectional view of a second support frame bar with a masking tape adhered to a second support surface of the second support frame bar, FIG. **17**A is a schematic structural view of a second support frame bar with a first protective film adhered to an inner edge of a second support surface of the second support frame bar, FIG. **17**B is a schematic sectional view of a second support frame bar with a second protective film adhered to an inner edge of a second support surface of the second support frame bar, FIG. **18**A is a schematic structural view of a second support frame bar with ultraviolet curing glue coated on a second support surface of the second support frame bar, FIG. **18**B is a schematic sectional view of a second support frame bar with ultraviolet curing glue coated on a second support surface of the second support frame bar, FIG. **19** is a schematic structural view when the masking tape is removed, and FIG. **20** is a schematic structural view of placing a support block in a region where the masking tape is removed. As shown in FIGS. **15**A to **20**, in an embodiment of the present disclosure, after the sub-backlight module is assembled, the process flow of fixing the display panel **51** to each second

support surface on the intermediate frame **11** is as follows:
(86) First, referring to FIGS. **15**A and **15**B, a first protective film **61** is adhered to an outer side wall of each second support frame bar of the intermediate frame **11**. The first protective film **61** is used to prevent the overflow glue from contacting the outer side wall of the second support frame bar when the glue overflows during the subsequent coating process of ultraviolet curing glue on the second support surface.
(87) Then, referring to FIGS. **16**A and **16**B, a masking tape **62** is adhered to the second support surfaces of the second support frame bars of the intermediate frame **11** at intervals. For example, three to five pieces of masking tape **62** with a width of 10 mm is adhered to each second support surface at intervals.
(88) Then, referring to FIGS. **17**A and **17**B, a second protective film **63** is adhered to an inner edge of the second support surface of the second support frame bar. The second protective film **63** may cover an edge of the optical film **18**. The second protective film **63** is used to prevent the overflow glue from contacting the optical film **18** in the sub-backlight module when the glue overflows during the subsequent coating process of ultraviolet curing glue on the second support surface.
(89) Then, referring to FIGS. **18**A and **18**B, ultraviolet curing glue **64** is continuously coated on the second support surface of the second support frame bar by a glue gun, where the ultraviolet curing glue **64** will cover a portion of the masking tape **62** on the second support surface. The glue is controlled to be coated with a width in the range of 0.6 mm to 0.8 mm, and a height in the range of 0.4 mm to 0.5 mm.
(90) Then, referring to FIG. **19**, the masking tape **62** is removed, and the ultraviolet curing glue **64** on the masking tape **62** is thus removed, so as to leave a notch **65** in the region originally covered by the masking tape.
(91) Next, referring to FIG. **20**, a support block **66**, which may be a foam having a width of 5 mm and a thickness of 2 mm, is placed in the notch **65**.
(92) Then, a double-sided tape is adhered to the second support surface of the first support frame bar (this step may be performed in advance), and the first protective film **61** and the second protective film **63** are removed.
(93) Then, the display panel **51** is placed on the support block **66**, and the display panel **51** is aligned with the sub-backlight module. At this time, the display panel **51** does not contact the ultraviolet curing glue **64**.
(94) Then, the ultraviolet curing glue **64** on each second support frame bar is cured in sequence, and the corresponding support block is removed. Specifically, the process of performing ultraviolet curing on a certain second support frame bar and removing the corresponding support block is as follows: while curing the ultraviolet curing glue **64** on the second support frame bar, the support blocks **66** on the second support frame bar are removed one by one, and the display panel is lightly touched and pressed to make the display panel in contact with the ultraviolet curing glue on the second support frame bar, until the ultraviolet curing glue is completely cured.
(95) Finally, the chip-on-film side region of the display panel is slightly touched and pressed (under 100 Kpa pressure for 30 seconds) so that a portion of the chip-on-film side region of the display panel is fixed with the double-sided tape.
(96) In some embodiments, in order to avoid light leakage from the side of the display panel and to shield the peripheral area of the display panel, the side and the peripheral area of the display panel **51** may be covered by a light-shielding tape **52**.
(97) Based on a same inventive concept, an embodiment of the present disclosure further provides a splicing display module. FIG. **21** is a schematic structural view of a splicing display module according to an embodiment of the present disclosure. As shown in FIG. **21**, in some embodiments, the splicing display module includes a display device, which may adopt a display substrate according to any of the above embodiments.
(98) In some embodiments, the peripheral area of the display device **71** includes: one chip-on-film

side region **81** and three non-chip-on-film side regions **82**. The chip-on-film side region **81** is arranged opposite to the second support surface on the first support frame bar, and the three non-chip-on-film side regions **82** are respectively arranged opposite to the second support surfaces on the three second support frame bars.

(99) The splicing display module specifically includes 2N display devices **71**. The 2N display devices **71** are arranged in an array of two rows and N columns, where N is a positive integer. Chip-on-film side regions **81** of N display devices **71** in a first row are all positioned on a side away from a second row of display devices **71**. Chip-on-film side regions **81** of N display devices **71** in the second row are all positioned on a side away from the first row of display devices **71**. The drawings merely exemplarily show a case where N is 2.

(100) In other words, for any two adjacent display devices **71**, the non-chip-on-film side regions **82** of the two display devices **71** form a seam. Taking each non-chip-on-film side region **82** having a width of 0.9 mm as an example, the non-chip-on-film side regions **82** of the two display devices **71** may form a seam with a width less than 2 mm.

(101) Taking the second row of display devices **71** in normal placement as an example, the first row of display devices **71** in this case are in an “upside down” state, so main boards **72** in the first row of display devices **71** are desired to have an image inversion function.

(102) The splicing display module provided in the embodiment of the present disclosure can implement extremely narrow splicing seams while realizing normal display of the splicing image.

(103) It will be appreciated that the above implementations are merely exemplary implementations for the purpose of illustrating the principle of the present disclosure, and the present disclosure is not limited thereto. It will be apparent to one of ordinary skill in the art that various modifications and variations may be made without departing from the spirit or essence of the present disclosure. Such modifications and variations should also be considered as falling into the protection scope of the present disclosure.

Claims

1. A support bracket for supporting a diffuser, characterized in comprising: a support post and a mounting part arranged in a first direction, wherein the mounting part comprises: an elastic connection part and a base arranged opposite to each other along the first direction, wherein the elastic connection part is connected to the base; wherein the support post extends in the first direction, and an end of the support post close to the mounting part is fixed to the elastic connection part; wherein the elastic connection part is configured to be elastically deformable in the first direction to change a distance between the end of the support post close to the mounting part and the base in the first direction; wherein the base is provided with a first screw hole; and wherein a portion of the base surrounding the first screw hole is outwardly convex toward a side away from the support post.

2. The support bracket according to claim 1, characterized in that the elastic connection part comprises: two elastic parts and a connection part arranged in a second direction, wherein the connection part is positioned between the two elastic parts, one end of each elastic part is connected to the base, the other end of the elastic part is connected to the connection part, and the end of the support post close to the mounting part is fixed to the connection part; wherein each elastic part is configured to be elastically deformable in the first direction to change a distance between the connection part and the base in the first direction, wherein the first direction is intersected with the second direction; wherein the elastic part is outwardly convex toward a side away from the base, and the connection part is arranged parallel to the base; wherein the elastic part comprises a first part, a second part and a third part connected in sequence; wherein the first part is connected to the base, and the third part is connected to the connection part; wherein from the connection between the second part and the first part to the connection between the second part and

the third part, a vertical distance between the second part and the base in the first direction gradually increases along the second direction; and wherein from the connection between the third part and the second part to the connection between the third part and the connection part, a vertical distance between the third part and the base in the first direction gradually decreases along the second direction.

3. The support bracket according to claim 1, characterized in that on the support post away from the mounting part in the first direction, the support post has a cross section perpendicular to the first direction with an area that first remains constant and then gradually decreases; and wherein the support post has the cross section perpendicular to the first direction with a rectangular or trapezoidal shape.

4. A sub-backlight module, characterized in comprising: a back plate comprising a bottom plate and a side plate; an intermediate frame and a diffuser, wherein the intermediate frame is assembled with and fixed to the back plate; and a support bracket for supporting the diffuser, wherein the support bracket comprises a support post and a mounting part arranged in a first direction, wherein the mounting part comprises a connection part and a base arranged opposite to each other along the first direction; wherein the connection part is connected to the base; wherein the support post extends in the first direction, and an end of the support post close to the mounting part is fixed to the connection part; wherein the side plate is positioned on a first side of the bottom plate and connected to an edge portion of the bottom plate; wherein the support bracket is positioned on the first side of bottom plate, with the mounting part of the support bracket fixed to the bottom plate; wherein the intermediate frame comprises: a first support structure and a second support structure, wherein a side of the first support structure away from the bottom plate is a first support surface, and the second support structure is positioned on a side of the first support surface away from the bottom plate; wherein the diffuser is positioned on the side of the first support surface away from the bottom plate and on an inner side of the second support structure, a clearance space is formed between the diffuser and an inner side wall of the second support structure, and an orthographic projection of the diffuser on the first support surface overlaps the first support surface; wherein a light guide bar is provided between the first support surface and the diffuser; and wherein a first side of the light guide bar close to the second support structure is in contact with the inner side wall of the second support structure.

5. The sub-backlight module according to claim 4, characterized in further comprising: a plurality of light bars arranged in a third direction, wherein each light bar comprises a plurality of lamp beads arranged in a fourth direction; and a ratio of the number of lamp beads to the number of support brackets is 6:1 to 10:1.

6. The sub-backlight module according to claim 5, characterized in that for any one of the support brackets, a center of an orthographic projection of the support post on the bottom plate overlaps with a center of a quadrangle enclosed by centers of orthographic projections of four lamp beads closest to the support bracket on the bottom plate.

7. The sub-backlight module according to claim 6, characterized in that a plurality of support brackets arranged in the fourth direction are provided on a central axis of the bottom plate, wherein the central axis of the bottom plate is a virtual line that passes through a center of a surface on a first side of the bottom plate and extends along the fourth direction.

8. The sub-backlight module according to claim 7, characterized in that the surface on the first side of the bottom plate has a first peripheral placement region, a central placement region and a second peripheral placement region sequentially arranged in the third direction; the first peripheral placement region and the second peripheral placement region are each provided with at least two support bracket sets, wherein the at least two support bracket sets comprise: a first support bracket set, and a second support bracket set on a side of the first support bracket set away from the central placement region, at least two light bars are provided between the first support bracket set and the second support bracket set; and the first support bracket set and the second support bracket set each

comprise at least two of the support brackets arranged in the fourth direction, and the number of support brackets in the first support bracket set is larger than the number of support brackets in the second support bracket set.
9. The sub-backlight module according to claim 5, characterized in that the base is provided with a first screw hole, and a reflective sheet is disposed on a side of the light bar facing away from the bottom plate; the reflective sheet is provided with a plurality of first openings in one-to-one correspondence with the lamp beads, and each lamp bead is exposed out of the corresponding first opening; the reflective sheet is provided with a plurality of second openings in one-to-one correspondence with the support brackets, and the bottom plate is provided with a plurality of second screw holes; and each support bracket is fixed to the bottom plate via a screw penetrating the corresponding second opening, the first screw hole, and the corresponding second screw hole in the bottom plate.
10. The sub-backlight module according to claim 5, characterized in that all the light bars are divided into a plurality of light bar sets, each light bar set comprises at least one light bar, each light bar set is configured with a corresponding adapter plate, and each light bar is electrically connected to the corresponding adapter plate through a first flexible flat cable; the adapter plate is fixed on a first side of the bottom plate, and the adapter plate is electrically connected to a power supply control module on a second side of the bottom plate through a second flexible flat cable penetrating the bottom plate; and a receiving slot is formed on the bottom plate in a region corresponding to the adapter plate, and the adapter plate is positioned in the corresponding receiving slot.
11. The sub-backlight module according to claim 4, characterized in that a second side of the light guide bar away from the second support structure exceeds an inner edge of the first support surface.
12. The sub-backlight module according to claim 11, characterized in that a prism film is formed on the inner side wall of the second support structure, and configured to reflect the light in the clearance space emitted to the inner side wall of the second support structure toward the light guide bar.
13. The sub-backlight module according to claim 11, characterized in that the intermediate frame comprises four support frame bars connected end to end in sequence, each support frame bar comprises the first support structure and the second support structure, and a side of the second support structure away from the bottom plate is a second support surface; the four support frame bars comprise one first support frame bar and three second support frame bars; and a ratio of a width of the second support surface on the first support frame bar to a width of the second support surface on each second support frame bar is 8:1 to 20:1.
14. A display device, characterized in comprising: the sub-backlight module as claimed in claim 4.
15. The display device according to claim 14, characterized in that the sub-backlight module further comprises: a intermediate frame and a diffuser, wherein the intermediate frame is assembled with and fixed to the back plate; the intermediate frame comprises: a first support structure and a second support structure, wherein a side of the first support structure away from the bottom plate is a first support surface, and the second support structure is positioned on a side of the first support surface away from the bottom plate; the diffuser is positioned on the side of the first support surface away from the bottom plate and on an inner side of the second support structure, a clearance space is formed between the diffuser and an inner side wall of the second support structure, and an orthographic projection of the diffuser on the first support surface overlaps the first support surface; a light guide bar is provided between the first support surface and the diffuser; a first side of the light guide bar close to the second support structure is in contact with the inner side wall of the second support structure; a second side of the light guide bar away from the second support structure exceeds an inner edge of the first support surface; the intermediate frame comprises four support frame bars connected end to end in sequence, each support frame bar comprises the first support structure and the second support structure, and a side of the second support structure away from the bottom plate is a second support surface; the four support frame

bars comprise one first support frame bar and three second support frame bars; a ratio of a width of the second support surface on the first support frame bar to a width of the second support surface on each second support frame bar is 8:1 to 20:1; wherein the display device further comprises: a display panel fixed to the second support surface; the display panel comprises an active area and a peripheral area surrounding the active area, and the peripheral area comprises: one chip-on-film side region and three non-chip-on-film side regions; and wherein the chip-on-film side region is arranged opposite to the second support surface on the first support frame bar, and the three non-chip-on-film side regions are respectively arranged opposite to the second support surfaces on the three second support frame bars.
16. A splicing display module, comprising: 2N display devices, which adopt the display device according to claim 15, wherein the 2N display devices are arranged in an array of two rows and N columns, where N is a positive integer; chip-on-film side regions of N display devices in a first row are all positioned on a side away from a second row of display devices; and chip-on-film side regions of N display devices in the second row are all positioned on a side away from the first row of display devices.
17. The sub-backlight module according to claim 4, characterized in that the connection part is elastic; and wherein the connection part is configured to be elastically deformable in the first direction to change a distance between the end of the support post close to the mounting part and the base in the first direction.
18. A sub-backlight module, characterized in comprising: a back plate comprising a bottom plate and a side plate; a support bracket for supporting a diffuser, wherein the support bracket comprises a support post and a mounting part arranged in a first direction, wherein the mounting part comprises a connection part and a base arranged opposite to each other along the first direction; and a plurality of light bars arranged in a third direction, wherein each light bar comprises a plurality of lamp beads arranged in a fourth direction, and a ratio of the number of lamp beads to the number of support brackets is 6:1 to 10:1; wherein the connection part is connected to the base; wherein the support post extends in the first direction, and an end of the support post close to the mounting part is fixed to the connection part; wherein the side plate is positioned on a first side of the bottom plate and connected to an edge portion of the bottom plate; wherein the support bracket is positioned on the first side of bottom plate, with the mounting part of the support bracket fixed to the bottom plate; wherein all the light bars are divided into a plurality of light bar sets, each light bar set comprises at least one light bar, each light bar set is configured with a corresponding adapter plate, and each light bar is electrically connected to the corresponding adapter plate through a first cable; wherein the adapter plate is fixed on a first side of the bottom plate, and the adapter plate is electrically connected to a power supply control module on a second side of the bottom plate through a second cable penetrating the bottom plate; and wherein a receiving slot is formed on the bottom plate in a region corresponding to the adapter plate, and the adapter plate is positioned in the corresponding receiving slot.
19. The sub-backlight module according to claim 18, characterized in that the connection part is elastic; and wherein the connection part is configured to be elastically deformable in the first direction to change a distance between the end of the support post close to the mounting part and the base in the first direction.
20. The sub-backlight module according to claim 18, characterized in that the first cable and the second cable are flexible flat.